

5G RAN

DRX Feature Parameter Description

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1 Change History

This chapter describes changes not included in the "Parameters", "Counters", "Glossary", and "Reference Documents" chapters. These changes include:

- Technical changes
Changes in functions and their corresponding parameters
- Editorial changes
Improvements or revisions to the documentation

1.1 5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 02 (2025-04-27).

- Technical changes

Change Description	Parameter Change	RAT	Base Station Model
Added an impact relationship between time-domain power saving BWP and DRX. For details, see 4.3.2.3 Function Impacts .	None	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Added the requirements of DRX for the setting of the NRDUCell.SlotAssignment parameter in high-frequency scenarios. For details, see 4.3.4 Cells .	None	High-frequency TDD	5900 series base stations (macro base stations)

- Editorial changes
Replaced the reserved-parameter bit that controls "PDCCH beam robustness for DRX in low-frequency NR TDD scenarios" with a formal-parameter option. For details, see [4.1.4.2 Switching Between Active Time and Sleep Time](#), [4.4.1.1 Data Preparation](#), and [4.4.1.2 Using MML Commands](#).
Revised other descriptions in this document, which do not involve changes in meanings and therefore are not detailed. Examples of such changes include changes in the names of reference documents.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Features in This Document

This document describes the following features.

Feature ID	Feature Name	Chapter/Section
FBFD-021103	DRX	4 DRX

2.3 Differences Between NSA and SA

Function Name	Difference	Chapter/Section
DRX	Supported in both NSA and SA networking, with the following differences: In NSA networking, the gNodeB does not instruct a UE in DRX mode to exit DRX mode when gap-assisted measurement is started. In SA networking, the gNodeB checks whether a UE in DRX mode needs to exit DRX mode based on actual conditions when gap-assisted measurement is started.	4 DRX

2.4 Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
DRX	Supported in both high and low frequency bands, with the following differences: Both SA networking and NSA networking are supported in low frequency bands, and NSA networking is supported in high frequency bands. SA networking is supported in high frequency bands only when DRX takes effect on inter-FR CA UEs. PDCCH beam robustness is supported only in low frequency bands. For details, see 4.1.4.2 Switching Between Active Time and Sleep Time.	4 DRX

3 Overview

With the ever increasing diversity of applications, terminal standby time has become a major concern for end users. Many smart-terminal applications use small sparse packets or heartbeat packets. These applications cause frequent communications between UEs and base stations. However, if UEs remain in RRC_CONNECTED mode and always monitor the physical downlink control channel (PDCCH), a lot of power will be consumed.

To reduce UE power consumption, 3GPP has introduced Discontinuous Reception (DRX) for NR and Huawei provides the 3GPP-compliant DRX feature accordingly. DRX enables UEs in connected mode to periodically suspend monitoring the PDCCH, thereby reducing the power consumption of UEs, especially when UEs are performing the following services:

- Periodic transmission of continuous small packets
- Delay-insensitive services
- Services characterized by small sparse packets

4 DRX

DRX is short for discontinuous reception. When the UE needs to receive downlink (DL) data or signaling, the UE turns on its receiver. In other situations, the UE turns off its receiver to reduce power consumption. This feature is controlled by the **BASIC_DRX_SW** option of the **NRDUCellUePwrSaving.NrDuCellDrxAlgoSwitch** parameter.

4.1 Principles

4.1.1 Basic Concepts

This section describes the basic concepts related to DRX.

On Duration

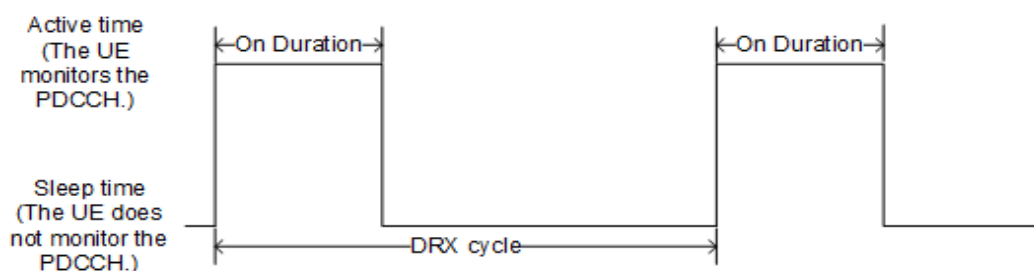
When DRX is used, a UE does not always keep its receiver off but periodically turns it on. Within a period of time after the DRX cycle starts, the UE monitors the PDCCH for incoming signaling. This period is called an On Duration and its length is defined by the On Duration Timer, which is specified by the **gNBDrxParamGroup.OnDurationTimer** parameter.

DRX Cycle

A DRX cycle is the interval between two occurrences of On Duration, as shown in [Figure 4-1](#).

A DRX cycle can be a **Long DRX Cycle** or **Short DRX Cycle**. Each DRX cycle is divided into **Active Time** and **Sleep Time** depending on whether the UE monitors the PDCCH.

Figure 4-1 DRX cycle



In a DRX cycle, after the On Duration period ends, if the UE switches from the active time to the sleep time (for details, see [4.1.4.2 Switching Between Active Time and Sleep Time](#)), the UE stays in the sleep time. Under the same condition, if the UE does not switch from the active time to the sleep time, the UE stays in the active time.

NOTE

An On Duration period is within the active time.

Active Time

In active time, the UE turns on its receiver and monitors the PDCCH.

According to section 5.7 "Discontinuous Reception (DRX)" in 3GPP TS 38.321 V15.6.0, a UE is in active time if one of the following conditions is met:

- The On Duration Timer, DRX Inactivity Timer, DL Retransmission Timer, UL Retransmission Timer, or Contention Resolution Timer is running. For details about the timers, see [DRX Timers](#).
- During the non-contention-based random access (RA), the UE has received a random access response but has not received the downlink control information (DCI) that indicates a new transmission.
- The UE has sent a scheduling request (SR) on the physical uplink control channel (PUCCH) and the SR is pending.

Sleep Time

Inactive time in a DRX cycle is called sleep time. During sleep time, the UE turns off its receiver.

Long DRX Cycle

Using a long DRX cycle can save more UE power. The long DRX cycle is specified by the **gNBDrxParamGroup.LongCycle** parameter.

Short DRX Cycle

A short DRX cycle delivers better service experience than a long DRX cycle but saves less power. The short DRX cycle is specified by the **gNBDrxParamGroup.ShortCycle** parameter.

By default, a long DRX cycle is configured for the UE. If a short DRX cycle is configured for the UE, the UE will determine when to use the long or short DRX

cycle based on the process described in [4.1.4.3 Switching Between Long and Short DRX Cycles](#).

DRX Timers

The following tables describe DRX timers.

 NOTE

The parameter in the table of a timer specifies the value of this timer.

Table 4-1 On Duration Timer

Item	Description
Parameter ID	gNBDUDrxParamGroup.OnDurationTimer
Function	This timer starts at the beginning of a DRX cycle. The UE monitors the PDCCH before the timer expires.
Start	This timer starts at the first subframe of a DRX cycle. For details, see 4.1.4.1 Starting a DRX Cycle .
Timing	Timing is based on milliseconds.
Stop	This timer stops upon expiry.
Expiry	Upon expiry, this timer stops timing.

Table 4-2 DRX Inactivity Timer

Item	Description
Parameter ID	In low-frequency bands, operators set the length of the DRX Inactivity Timer (specified by the gNBDUDrxParamGroup.DrxInactivityTimer parameter) to a small value considering UE power saving. In inter-gNodeB CA scenarios, if the inter-gNodeB delay is longer than the length of the DRX Inactivity Timer, the gains of inter-gNodeB CA decrease. To address this issue, Huawei introduces the inactivity timer length adaptation function. For details about inter-gNodeB CA (including fast inter-gNodeB CA and inter-gNodeB delay), see <i>CA Experience Improvement</i>. The inactivity timer length adaptation function is controlled by the NRDUCellUePwrSaving.IAdaptiveAlgoSwitch parameter. • If the NRDUCellUePwrSaving.IAdaptiveAlgoSwitch parameter is set to OFF, the inactivity timer length adaptation function is disabled in inter-gNodeB CA scenarios, and the DRX Inactivity Timer length takes the value of the gNBDUDrxParamGroup.DrxInactivityTimer parameter. • If the NRDUCellUePwrSaving.IAdaptiveAlgoSwitch parameter is set to ON_IPRAN_CA, the inactivity timer length adaptation function is enabled in inter-gNodeB CA scenarios. In this case, the gNodeB automatically generates the length of the DRX Inactivity Timer based on the activation status of inter-gNodeB CA, the gNBDUDrxParamGroup.DrxInactivityTimer parameter value, and inter-gNodeB delay in inter-gNodeB CA scenarios. – The inactivity timer length adaptation function in inter-gNodeB CA scenarios can take effect only on UEs with inter-gNodeB CA in effect. The DRX Inactivity Timer length may not take the value of the gNBDUDrxParamGroup.DrxInactivityTimer parameter for these UEs. – The inactivity timer length adaptation function in inter-gNodeB CA scenarios cannot take effect on UEs without inter-gNodeB CA in effect. The DRX Inactivity Timer length takes the value of the gNBDUDrxParamGroup.DrxInactivityTimer parameter for these UEs. In high-frequency bands, the inactivity timer length takes the value of the gNBDUDrxParamGroup.DrxInactivityTimer parameter.
Function	This timer is used for a UE to determine whether to extend the active time because of the arrival of new data. It also provides a reference for the UE to determine when to use a short DRX cycle.
Start	This timer starts or restarts when the UE successfully decodes a PDCCH that indicates a new UL or DL transmission (UL grant or DL data indication) for this UE.

Item	Description
Timing	Timing is based on milliseconds.
Stop	This timer stops upon expiry.
Expiry	Upon expiry, this timer stops timing and one of the following operations is performed: • If the UE is in a long DRX cycle: – If a short DRX cycle has been configured, the Short Cycle Timer starts and the short DRX cycle starts. – If no short DRX cycle is configured, the long DRX cycle continues. • If the UE is in a short DRX cycle, the Short Cycle Timer restarts.

Table 4-3 Short Cycle Timer

Item	Description
Parameter ID	gNBDrxParamGroup.ShortCycleTimer
Function	This timer specifies the lifetime of a short DRX cycle, that is, the number of consecutive repetitions of a short DRX cycle. If the gNBDrxParamGroup.ShortCycleTimer parameter is set to 0 , the gNBDrxParamGroup.ShortCycle parameter does not take effect.
Start	After the DRX Inactivity Timer expires, the Short Cycle Timer starts or restarts depending on the following: • If the UE is in a long DRX cycle: – If a short DRX cycle has been configured, the short DRX cycle starts and the Short Cycle Timer starts. – If no short DRX cycle is configured, the long DRX cycle continues and the Short Cycle Timer does not start. • If the UE is in a short DRX cycle, the Short Cycle Timer restarts.
Timing	Timing is based on the number of repetitions of a short DRX cycle.
Stop	This timer stops upon expiry.
Expiry	Upon expiry, this timer stops timing and the UE starts to use a long DRX cycle.

Table 4-4 DL Retransmission Timer

Item	Description
Parameter ID	gNBDUDrxParamGroup.DrxRetransTimer
Function	This timer specifies the maximum period of time the UE waits for downlink retransmission in active time. NOTE If the UE has not received the downlink retransmitted data before this timer expires, the UE will no longer accept it.
Start	This timer starts or restarts when the DL HARQ-RTT Timer expires and the UE does not decode the corresponding downlink data correctly.
Timing	Timing is based on the number of slots.
Stop	• This timer stops when the DL HARQ-RTT Timer starts. • This timer stops upon expiry.
Expiry	Upon expiry, this timer stops timing.

Table 4-5 UL Retransmission Timer

Item	Description
Parameter ID	gNBDUDrxParamGroup.DrxRetransTimer
Function	This timer specifies the maximum period of time the UE waits for uplink retransmission in active time. NOTE If the UE has not received the uplink grant before this timer expires, the UE considers that retransmission is no longer needed.
Start	This timer starts or restarts when the UL HARQ-RTT Timer expires.
Timing	Timing is based on the number of slots.
Stop	• This timer stops when the UL HARQ-RTT Timer starts. • This timer stops upon expiry.
Expiry	Upon expiry, this timer stops timing.

Table 4-6 DL HARQ-RTT Timer

Item	Description
Parameter ID	N/A (The timer length is fixed to 56 OFDM symbols.)

Item	Description
Function	This timer specifies the minimum interval between the time when the gNodeB receives the HARQ feedback of a downlink data transmission and the time when the downlink data retransmission starts. This timer is used to determine when to start the DL Retransmission Timer.
Start	This timer starts when the UE returns the ACK/NACK for a downlink data transmission on the PUCCH.
Timing	Timing is based on the number of OFDM symbols.
Stop	This timer stops upon expiry.
Expiry	Upon expiry, this timer stops timing. If the UE does not decode the corresponding downlink data correctly, the DL Retransmission Timer starts.

Table 4-7 UL HARQ-RTT Timer

Item	Description
Parameter ID	N/A (The timer length is fixed to 56 OFDM symbols.)
Function	This timer specifies the minimum interval between the time when the UE completes data transmission on the PUSCH and the time when the UE receives the next UL Grant on the PDCCH. This timer is used to determine when to start the UL Retransmission Timer.
Start	This timer starts or restarts each time data transmission on the PUSCH completes.
Timing	Timing is based on the number of OFDM symbols.
Stop	This timer stops upon expiry.
Expiry	Upon expiry, this timer stops timing and the UL Retransmission Timer starts.

Table 4-8 Contention Resolution Timer

Item	Description
Parameter ID	N/A (The timer length is fixed to 64 subframes.)
Function	This timer specifies the time during which the UE waits for a contention resolution in a contention-based random access procedure.

Item	Description
Start	This timer starts when the UE initially transmits or retransmits an Msg3 in a contention-based random access procedure.
Timing	Timing is based on the number of subframes.
Stop	This timer stops when the UE receives a contention resolution message.
Expiry	Upon expiry, this timer stops timing and the UE retransmits a preamble.

4.1.2 QCI- and RFSP-based DRX Parameter Configuration

QCI-based DRX Parameter Configuration

Quality of service (QoS) is a measure that helps define the level of service and provides an indication of customer satisfaction. The QoS class identifier (QCI) is used to identify the QoS class of a data radio bearer (DRB). QCI-based DRX parameter configuration allows different DRX parameter groups to be configured for different QCIs. The **NRCellQciBearer** and **NRDUCellQciBearer** MOs are used to bind a QCI to a DRX parameter group (configured using the **gNBDrxParamGroup** and **gNBDUDrxParamGroup** MOs).

DRX takes effect on a UE only when all of the following conditions are met:

- The QCIs added to the **NRCellQciBearer** and **NRDUCellQciBearer** MOs must exist in the **gNBQciBearer** MO.
- All QCIs used by the UE are bound to valid DRX parameter groups. That is, neither **NRCellQciBearer.DrxParamGroupId** nor **NRDUCellQciBearer.DuDrxParamGroupId** is set to 255.

A UE may have multiple radio bearers, each corresponding to a QCI. Different QCIs have different priorities (specified by the **gNBQciBearer.PriorityLevel** parameter). After DRX takes effect on the UE, if the UE is not running services carried on bearers with QCI 1, 65, or 66, the UE uses the DRX parameter group bound with the QCI of the highest priority.

 NOTE

In SA networking, the core network, that is 5GC, notifies the gNodeB of the 5G QoS Identifier (5QI) corresponding to each QoS flow. For detailed description about QCI and 5QI, see *QoS Management*.

Each QCI is bound to a set of DRX parameter groups. Different QCIs can be bound to the same set of DRX parameter groups.

DRX is not recommended for QCI 1, QCI 65, QCI 66, and QCI 69. Therefore, DRX parameter groups are not added or bound for such QCIs by default. If there are DRX parameter groups for QCI 1, QCI 65, or QCI 66, the DRX parameter group is selected for a UE by QCI priority in the following sequence: QCI 65 > QCI 1 > QCI 66 > other QCIs.

For QCI 2, QCI 3, QCI 5, QCI 7, QCI 75, QCI 79, QCI 4, QCI 6, QCI 8, QCI 9, and QCI 70, the system adds and binds DRX parameter groups by default. DRX parameter groups must be manually added and bound for QCIs other than the preceding ones.

The default DRX parameter groups bound to QCIs vary depending on whether a cell is a high- or low-frequency cell.

RFSP-based DRX Parameter Configuration

An operator registers RAT/frequency selection priorities (RFSPs) (ranging from 1 to 256) for UEs with the unified data management (UDM) database. In this way, gNodeBs can perform RFSP-based special processing for services performed by certain UEs and provide UE-specific services based on UEs' service characteristics and loading status. RFSP-based DRX parameter configuration is to configure different DRX parameter groups for different RFSPs. Specifically, RFSPs are bound to DRX parameter groups (specified in the **gNBDrxParamGroup** and **gNBDUDrxParamGroup** MOs) using the **gNBRfspConfig** and **gNBDuRfspConfig** MOs. For details, see *Flexible User Steering*.

If both QCI bearers and RFSPs are configured for UEs, the gNodeB preferentially configures DRX parameter groups based on RFSPs.

4.1.3 Entering and Exiting DRX Mode

As shown in [Figure 4-2](#), in a cell where the DRX feature is enabled (by selecting the **BASIC_DRX_SW** option of the **NRDUCellUePwrSaving.NrDuCellDrxAlgoSwitch** parameter), the base station performs the following operations for a UE each time DRX is triggered (see [Triggering Scenarios](#)).

- For a UE not in DRX mode, the base station checks whether the UE meets conditions for entering DRX mode (see [Entering Conditions](#)):
 - If the conditions are met, the base station sends the UE the DRX configuration. That is, the base station configures the *DRX-Config* field in an RRCConnectionReconfiguration message (NSA networking) or an RRCReconfiguration message (SA networking) and then sends this message to the UE. The UE then enters DRX mode based on parameter settings in this field.
 - If the conditions are not met, the UE stays in non-DRX mode.
- For a UE in DRX mode, the base station checks whether the UE meets the DRX exiting conditions (see [Exiting Conditions](#)):
 - If the DRX exiting conditions are met, the base station sends the UE an RRCConnectionReconfiguration message (in NSA networking) or an

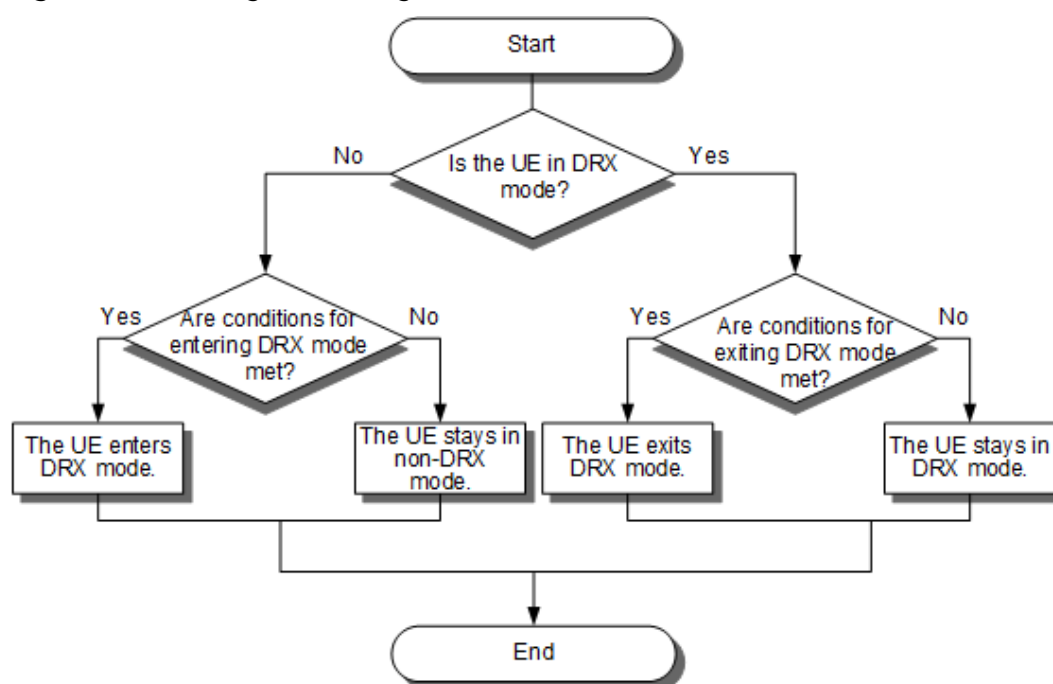
RRCReconfiguration message (in SA networking) with the *DRX-Config* field unspecified. After receiving the message, the UE immediately clears all the DRX parameters stored locally and exits DRX mode.

- If the DRX exiting conditions are not met, the UE stays in DRX mode.

NOTE

If the UE has multiple radio bearers with different QCI, the DRX parameter group bound with the QCI of the highest priority is used. If the radio bearer with the highest QCI priority changes, DRX parameter settings are updated when the UE stays in DRX mode.

Figure 4-2 Entering and exiting DRX mode



Triggering Scenarios

- The UE initially accesses the network.
- The UE performs handovers.
- The base station creates or deletes a radio bearer for the UE.
- The base station configures measurement gaps for the UE.
- In carrier aggregation (CA) scenarios, the base station adds or removes a secondary component carrier (SCC) for the UE.
- In low-frequency scenarios, the base station delivers the periodic sounding reference signal (SRS) configuration of the UE.

Entering Conditions

A UE enters DRX mode when all the following conditions are met:

- The DRX feature is enabled for cells in common scenarios. In CA scenarios, the DRX feature is enabled for both the PCell and SCell of the UE.

- The UE reports that DRX is supported, that is, the *MAC-ParametersXDD-Diff* IE contains the **longDRX-Cycle** field. For details, see section 6.3.3 "UE capability information elements" in 3GPP TS 38.331 V15.6.0.
- All radio bearers set up for the UE support the DRX feature, that is, DRX parameter groups have been configured for QCI of these radio bearers and the DRX parameter group IDs are valid (neither **NRCellQciBearer.DrxParamGroupId** nor **NRDUCellQciBearer.DuDrxParamGroupId** is set to 255).
- Long DRX cycle (specified by **gNBDrxParamGroup.LongCycle**) $\leq \{TATimer\text{ length (NRDUCellUtaConfig.UlTimeAlignmentTimer)} \times 1/6\}$

Even if all the preceding conditions are met, the UE does not immediately enter DRX mode in the following situations:

- In SA networking, if a UE does not meet the conditions for coexistence of DRX and gap-assisted measurement, the gNodeB does not instruct the UE to enter DRX mode when the UE is performing continuous gap-assisted measurements. DRX and gap-assisted measurement can coexist when all of the following conditions are met:
 - Coexistence of DRX and gap-assisted measurement (controlled by the **DRX_AND_GAP_MEAS_COEXIST_SW** option of the **NRDUCellUePwrSaving.NrDuCellDrxAlgoSwitch** parameter) is enabled. This function takes effect only for low-frequency cells.
 - The length of the long DRX cycle of the UE (specified by **gNBDrxParamGroup.LongCycle**) is greater than or equal to 80 ms.
 - The length of the DRX On Duration Timer of the UE (specified by **gNBDrxParamGroup.OnDurationTimer**) is greater than or equal to 20 ms. Alternatively, the DRX On Duration Timer does not overlap with the gap-assisted measurement in the time domain.
- If DRX exit of UEs performing full-buffer services is enabled in a cell (with the **FULL_BUFFER_UE_EXIT_DRX_SW** option of the **NRDUCellUePwrSaving.NrDuCellDrxAlgoSwitch** parameter selected) and the UE is identified as a UE performing full-buffer services, the UE does not enter the DRX mode. When the UE is no longer identified as a UE performing full-buffer services, the gNodeB instructs the UE to enter DRX mode through an RRC connection reconfiguration message.
- In low-frequency TDD scenarios, downlink enhancement for power saving is enabled in a cell (by selecting the **PWR_SAVING_DOWNLINK_ENH_SW** option of the **NRDUCellPdsch.DlLinkAdaptEnhancementSw** parameter), service performance is preferentially guaranteed for UEs that initially access the network and DRX takes effect after a delay. That is, the gNodeB does not instruct the UE to enter DRX mode before the first SRS weight measurement is complete. After the UE completes the first SRS weight measurement, the gNodeB instructs the UE to enter DRX mode through the RRCReconfiguration message. This function takes effect only for low-frequency TDD cells served by the MetaAAU.

 NOTE

- The UE can perform reportCGI measurements only after entering the DRX sleep time. Therefore, when the gNodeB instructs the UE to report the NCGI or ECGI, the UE enters the DRX mode.
- For UEs in RRC_CONNECTED mode, DRX-related parameter modifications take effect only upon the next RRC connection reconfiguration.

Exiting Conditions

The gNodeB instructs the UE to exit DRX mode when any of the following conditions is met:

- If DRX is disabled in a cell in common scenarios, the gNodeB instructs the UE to exit DRX mode during the next RRC connection reconfiguration procedure. In CA scenarios, if DRX is disabled in either the PCell or SCell of a UE, the gNodeB instructs the UE to exit DRX mode during the next RRC connection reconfiguration procedure. (If an SCC that does not support DRX is added for the UE, the RRC connection reconfiguration is triggered immediately.)
- The UE has set up a radio bearer (for example, a delay-sensitive radio bearer) for which DRX does not take effect. That is, the QCI of such a radio bearer is not bound to any DRX parameter group or is bound to a parameter group whose ID is 255 (**NRCellQciBearer.DrxParamGroupId** or **NRDUCellQciBearer.DuDrxParamGroupId** is set to 255).
- Long DRX cycle (specified by **gNBDrxParamGroup.LongCycle**) > {TATimer length (**NRDUCellUtaConfig.ULTimeAlignmentTimer**) × 1/6}

In SA networking, if a UE does not meet the conditions for coexistence of DRX and gap-assisted measurement, the gNodeB instructs the UE in DRX mode to exit DRX mode when gap-assisted measurement is started. Otherwise, the gNodeB does not instruct the UE in DRX mode to exit DRX mode.

In NSA networking, the gNodeB does not instruct the UE to exit DRX mode when gap-assisted measurement is started.

After DRX is enabled, in weak coverage scenarios, the DRX timer status (active time or sleep time) of a UE may be inconsistent with that on the base station side, causing service drops, increased service delay, and decreased user-perceived rates. To prevent this, exit of weak-coverage UEs from DRX is introduced. This function is controlled by the **WEAK_COV_DRX_EXIT_SW** option of the **NRDUCellUePwrSaving.NrDuCellDrxAlgoSwitch** parameter. When this option is selected, the processing mechanism on the base station is as follows:

- If a UE initially accesses a cell and meets any of the following conditions, the UE is identified as a weak-coverage UE that initially accesses the network in UE power saving scenarios. For such a UE, DRX parameters will not be included in the RRC reconfiguration message.
 - The Msg3-based DMRS RSRP is less than the sum of values of **NRDUCellDmrs.InitDmrsRsrpWeakCovThld** and **NRDUCellUePwrSaving.WeakCoverageThldOffset**.
 - The Msg3-based DMRS SINR is less than the sum of values of **NRDUCellDmrs.InitDmrsSinrWeakCovThld** and **NRDUCellUePwrSaving.WeakCoverageThldOffset**.

- If DRX parameters are configured for a UE through RRC signaling and any of the following conditions is met, the UE is identified as a weak-coverage UE in UE power saving scenarios. Such a UE will exit DRX mode.
 - The CQI measurement result periodically reported by the UE is less than the value of **NRDUCellUePwrSaving.WeakCoverageCqiThld**. The **NRDUCellUePwrSaving.WeakCoverageCqiThld** parameter is designed based on the 64QAM-specific mapping between CQI and modulation schemes. If 256QAM is used, the gNodeB converts the parameter value based on the same spectral efficiency and then compares the converted value with the CQI measurement result reported by the UE.
 - The uplink SINR is less than the value of **NRDUCellUePwrSaving.WeakCoverageULSinrThld**.

4.1.4 Running DRX Mode

After a UE enters DRX mode, a DRX cycle will start (see [4.1.4.1 Starting a DRX Cycle](#)) if certain conditions are met, and the base station will switch the UE between active time and sleep time (see [4.1.4.2 Switching Between Active Time and Sleep Time](#)) and between a long DRX cycle and a short DRX cycle (see [4.1.4.3 Switching Between Long and Short DRX Cycles](#)) based on network environments and parameter configurations.

4.1.4.1 Starting a DRX Cycle

Upon entering DRX mode, a UE does not immediately start the On Duration Timer for a DRX cycle. The On Duration Timer starts, which means, the UE starts a DRX cycle, in the following scenarios:

- A long DRX cycle is started when the following conditions are met:

$$[(SFN \times 10) + SSFN] \bmod (gNBDrxParamGroup.LongCycle) = DRX\ Start\ Offset$$
- A short DRX cycle is started when the following conditions are met:
 - The value of **gNBDrxParamGroup.ShortCycleTimer** is not 0.
 - $[(SFN \times 10) + SSFN] \bmod (gNBDrxParamGroup.ShortCycle) = (DRX\ Start\ Offset) \bmod (gNBDrxParamGroup.ShortCycle)$

In the preceding formulas:

- SFN: system frame number
- SSFN: system subframe number
- modulo (x): The modulo operation is performed on x.
- **gNBDrxParamGroup.LongCycle** and **gNBDrxParamGroup.ShortCycle** specify the durations of the long and short DRX cycles, respectively. The length of the long DRX cycle is *K* times of that of the short DRX cycle. *K* is an integer greater than or equal to 2.
- *DRX Start Offset* is the start offset of the long DRX cycle, and $(DRX\ Start\ Offset) \bmod (gNBDrxParamGroup.ShortCycle)$ is the start offset of the short DRX cycle.
- The base station notifies the UE of the cycle duration and *DRX Start Offset* using the RRCConnectionReconfiguration message (in NSA networking) or the RRCReconfiguration message (in SA networking).

NOTE

If a UE reports that it supports long DRX cycles, that is, the *MAC-ParametersXDD-Diff* IE contains the **longDRX-Cycle** field, the UE can use a long DRX cycle. If a UE reports that it supports short DRX cycles, that is, the *MAC-ParametersXDD-Diff* IE contains the **shortDRX-Cycle** field, the UE can use a short DRX cycle. For details, see section 6.3.3 "UE capability information elements" in 3GPP TS 38.331 V15.6.0.

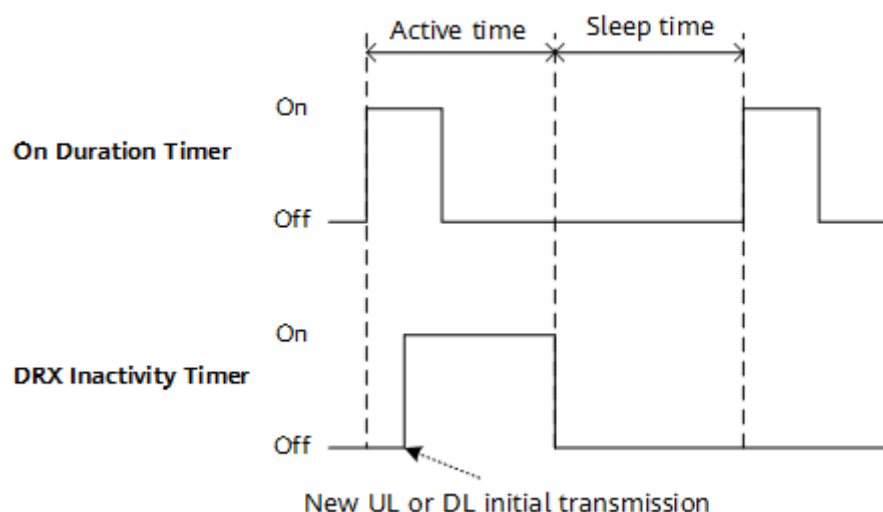
4.1.4.2 Switching Between Active Time and Sleep Time

Switching between active time and sleep time depends on DRX timers and other service processes.

The following describes how the UE receiver switches between the active time and sleep time based on different timers and other service processes. For details about the timer running mechanisms, see [DRX Timers](#).

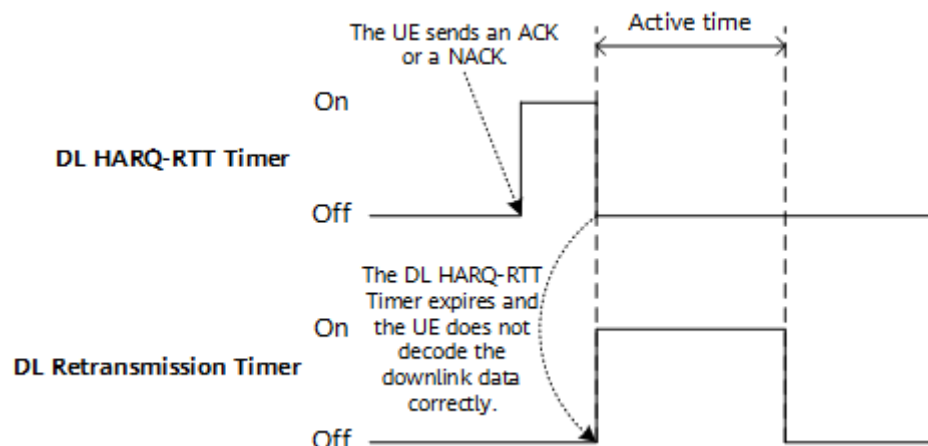
- When the On Duration Timer or DRX Inactivity Timer is running, the UE is in the active time, as shown in [Figure 4-3](#).

Figure 4-3 Active time determined by the On Duration Timer and DRX Inactivity Timer



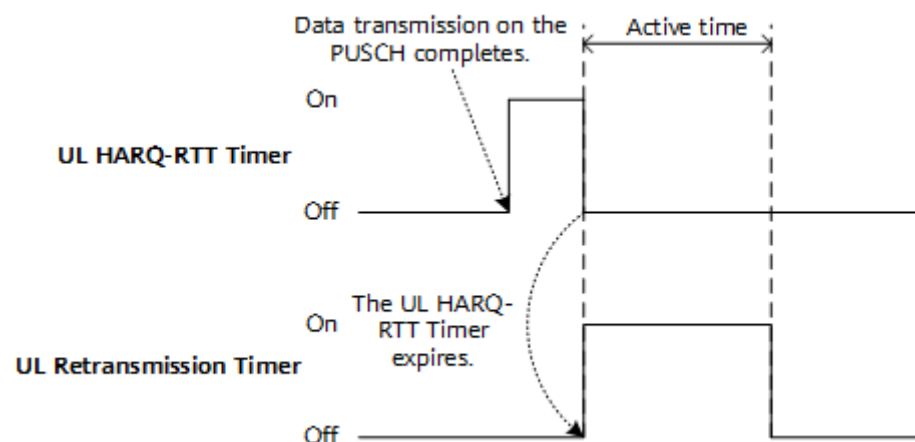
- When the DL HARQ-RTT Timer expires and the UE does not correctly decode the corresponding downlink data, the DL Retransmission Timer is triggered. During the running of the DL Retransmission Timer, the UE is in the active time, as shown in [Figure 4-4](#).

Figure 4-4 Active time determined by the DL HARQ-RTT Timer and DL Retransmission Timer



- When the UL HARQ-RTT Timer expires, the UL Retransmission Timer starts. During the running of the UL Retransmission Timer, the UE is in active time, as shown in [Figure 4-5](#). For details about the periods when the UE is in DRX active time, see [Active Time](#).

Figure 4-5 Active time determined by the UL HARQ-RTT Timer and UL Retransmission Timer



The gNodeB and UE use their own DRX timers. That is, the gNodeB determines whether a UE is in the active or sleep time based on the timers on the gNodeB. When the gNodeB mistakenly detects SRs from a UE in the DRX sleep time or the UE in the DRX active time fail to detect DCI from the gNodeB, the DRX Inactivity Timer and Short Cycle Timer are inconsistent on the gNodeB and UE sides. As a result, the gNodeB may determine that the UE is in the DRX active time while it is actually in the DRX sleep time, which causes the following impacts:

- The UE does not respond to the scheduling command from the gNodeB, and the scheduling result is discontinuous transmission (DTX). This increases the block error rate (BLER) over the air interface and the number of used PDCCH control channel elements (CCEs).
- UE's measurement results (such as the SRS and CSI-RS feedback) are inaccurate.

To address this issue, the gNodeB checks the DRX state of UEs based on the scheduling feedback.

- In the uplink, if the **UL_BASED_DRX_STATUS_DETECT_SW** option of the **NRDUCellUePwrSaving.NrDuCellDrxAlgoSwitch** parameter is selected, DRX status evaluation based on scheduling feedback over the PUSCH takes effect, and the gNodeB starts DRX status evaluation based on uplink scheduling feedback.
- In the downlink, the gNodeB starts DRX status evaluation based on downlink scheduling feedback by default.

After DRX status evaluation based on scheduling feedback is started, the gNodeB determines that a UE is in the DRX sleep time and therefore stops the DRX Inactivity Timer and Short Cycle Timer if any of the following conditions is met in the uplink or downlink:

- When the gNodeB detects an SR from the UE in the DRX sleep time, the number of consecutive DTXs in the uplink or downlink scheduling feedback is greater than or equal to the DRX sleep time detection threshold. The DRX sleep time detection threshold is specified by the **NRDUCellUePwrSaving.DrxSleepStateDetectThld** parameter.
- In other scenarios, the number of consecutive DTXs in the uplink or downlink scheduling feedback is twice the DRX sleep time detection threshold or greater.

In low-frequency NR TDD scenarios, if DRX status evaluation based on scheduling feedback is enabled and missing DCI detections occur on a UE that is in the DRX active time, the gNodeB may still incorrectly determine the DRX status of a UE. In this context, PDCCH beam robustness is introduced. This function is controlled by the **PDCCH_BEAM_ROBUST_SW** option of the **NRDUCellPdcch.PdcchAlgoSwitch** parameter. LampSite base stations do not support this function. When this option is selected, this function takes effect. If the gNodeB detects that the number of consecutive DTXs in the uplink or downlink scheduling feedback is greater than or equal to twice the DRX sleep time detection threshold during the DRX active time, the gNodeB switches the PDCCH beam selection method and determines the DRX status of the UE based on the post-switching consecutive feedback of uplink or downlink scheduling.

- If the consecutive feedback of uplink or downlink scheduling is not DTXs after the switching, the gNodeB determines that the UE is in the DRX active time and restarts the DRX Inactivity Timer.
- If the consecutive feedback of uplink or downlink scheduling is DTXs after the switching, the gNodeB determines that the UE is in the DRX sleep time and stops the DRX Inactivity Timer and Short Cycle Timer.

NOTE

PDCCH beam selection can be based on SRS beam measurement or CSI-RS measurement. For details, see section "Management of Control Beams" in *Beam Management*.

4.1.4.3 Switching Between Long and Short DRX Cycles

The gNodeB delivers parameters of both long and short DRX cycles to the UE, but the UE uses only a long or short DRX cycle at a time. The UE can switch between the long and short DRX cycles according to certain rules. The duration of the long

DRX cycle is specified by the **gNBDrxParamGroup.LongCycle** parameter, and the duration of the short DRX cycle is specified by the **gNBDrxParamGroup.ShortCycle** parameter.

- Switching from a long DRX cycle to a short DRX cycle
By default, a UE uses a long DRX cycle. If the base station has configured a short DRX cycle for the UE, the UE switches from the long DRX cycle to the short DRX cycle when the DRX Inactivity Timer expires. In this way, service delay is shortened.
- Switching from a short DRX cycle to a long DRX cycle
When the Short Cycle Timer expires, the UE exits the short DRX cycle and enters a long DRX cycle, reducing UE power consumption.

 NOTE

The Short Cycle Timer specifies the number of consecutive repetitions of a short DRX cycle. When the UE enters a short DRX cycle, the Short Cycle Timer immediately starts. When the Short Cycle Timer expires, the UE exits the short DRX cycle and enters a long DRX cycle.

In high-frequency scenarios, UEs cannot enter the long DRX cycle after entering the short DRX cycle. In addition, if the **On Duration Timer**, **DRX Inactivity Timer**, or **Short Cycle** parameter is set improperly, UEs cannot enter the sleep time for power saving. Therefore, it is recommended that short DRX cycles not be configured in high-frequency scenarios. If a short DRX cycle is required, the configurations for the default bearers (QCI 2, QCI 3, QCI 7, QCI 67, QCI 75 and QCI 79) must be modified and all the following conditions must be met:

- The value of **gNBDrxParamGroup.ShortCycle** is greater than that of **gNBDUDrxParamGroup.OnDurationTimer**.
- The **gNBDrxParamGroup.ShortCycleTimer** parameter is set to a value other than 0.

4.2 Network Analysis

4.2.1 Benefits

DRX offers the following advantages over continuous reception:

- A UE does not need to constantly monitor the PDCCH. Instead, the UE turns on its receiver only when necessary. This reduces UE power consumption and ensures a longer standby time.
- In a low-frequency cell, cell global identifier (CGI) measurement for ANR is allowed for the UE in the DRX sleep time.

After DRX is enabled, the power saving effect of UEs can be monitored by using **N.Cdrx.Sleep.Dur.Total** (total duration for RRC_CONNECTED DRX UEs in sleep time in a cell) and **N.Cdrx.Active.Dur.Total** (total duration for RRC_CONNECTED DRX UEs in active time in a cell).

 NOTE

When a UE is in DRX mode, its power consumption can be reduced. The power saving effect is related to UE chips. Therefore, using the same DRX parameters may not bring the same power saving effect for different UEs.

4.2.2 Impacts

The impacts between this function and other functions are described in [4.3.2.3 Function Impacts](#).

DRX introduces the sleep time, which reduces UE power consumption and has the following impacts on the network:

- The measurement and reporting periods of the channel state information (CSI) and the synchronization signal and PBCH block (SSB) increase and the rank adjustment slows, which affects the delay and throughput. In addition, the handover success rate and SCG addition success rate decrease, and the service drop rate and HARQ retransmission rate increase. If **gNBDrxParamGroup.LongCycle** is set to an excessively large value, or UEs are moving at high speeds or under weak coverage, the preceding impacts will be more evident. Longer SSB measurement and reporting periods may result in fewer RRM measurements.
- The CQI measurement and reporting periods are prolonged. Therefore, the measured CQI value will fluctuate depending on the channel environment.
- The number of times UEs are scheduled may decrease.
- The maximum RLC retransmission interval is prolonged. As a result, the proportion of successful RLC retransmissions may increase, the number of abnormal UE context releases in a cell (indicated by **N.UECntx.AbnormRel**) may decrease, and the service drop rate may decrease.
- The consumption of PDCCH CCEs, PDSCH PRBs, and PUSCH PRBs increases. There is a false detection probability when the gNodeB detects whether a UE sends SRs. Specifically, although a UE does not send an SR, the gNodeB detects the SR. When such false detection occurs in the DRX sleep time, the gNodeB mistakenly considers that the UE is in the DRX active time and performs scheduling on the UE. However, the UE is actually in the sleep time and does not respond to the scheduling. This increases the BLER and retransmissions from the gNodeB, raising up the consumption of CCEs and PRBs.
- After DRX status evaluation based on scheduling feedback is enabled, the number of scheduled UEs and the cell throughput increase in heavily loaded cells. However, false SR detection may be incorrectly identified, causing the UE throughput to decrease.
- For low-frequency cells, this feature may have the following impacts:
 - The service delay of the UE increases, and thereby the average downlink UE throughput (indicated by **User Downlink Average Throughput (DU)**) decreases.
 - The number of UEs with data in the buffer increases, and the number of activated UEs increases.
 - If the sleep time is too short, CGI measurements for ANR will fail.
- For high-frequency cells:
 - The number of uplink aperiodic CSI reporting times and the number of SRS preallocation times triggered by DCI for uplink scheduling decrease, reducing the number of padding packets. As a result, the total volume of uplink data received at the MAC layer in a cell (indicated by **N.ThpVol.UL.Cell**) may decrease.

- The average uplink UE throughput (**User Uplink Average Throughput (DU)**), uplink RLC throughput distribution (**N.Thp.UL.Samp.Index0** to **N.Thp.UL.Samp.Index14**), and average uplink cell throughput (**Cell Uplink Average Throughput (DU)**) fluctuate. The fluctuation is related to the user service type, user distribution and mobility, weak coverage proportion, and user scheduling behavior change caused by the decrease in padding packets.
- In scenarios where UEs performing small-packet services account for a large proportion, DRX may increase the service delay of UEs. As a result, the average downlink UE throughput (indicated by **User Downlink Average Throughput (DU)**) may decrease, the number of UEs with data in the buffer may increase, and the number of activated UEs may increase.
- In scenarios where UEs performing large-packet services account for a large proportion, DRX may increase the average downlink UE throughput (indicated by **User Downlink Average Throughput (DU)**) and decrease the number of UEs with data in the buffer and the number of activated UEs. This is because UEs performing small-packet services are more likely to enter the sleep time and the probability that these UEs occupy analog beam resources decreases, reducing the preemption on analog beam resources for UEs performing large-packet services and the scheduling duration of UEs performing large-packet services.

In heavy-traffic scenarios, the preceding impacts of DRX increase. After DRX is enabled for high-frequency cells, the preceding impacts increase because the aperiodic scheduling of resources such as CSI-RS and SRS is affected by DRX.

For low-frequency TDD cells in light-load scenarios with small packets transmitted in the uplink, the basic K2 values of UEs decrease. This shortens the uplink retransmission delay and enables UEs to enter the DRX sleep time more quickly. The value of **N.Cdrx.Sleep.Dur.Total** (total duration for RRC_CONNECTED DRX UEs in sleep time in a cell) increases and that of **N.Cdrx.Active.Dur.Total** (total duration for RRC_CONNECTED DRX UEs in active time in a cell) decreases. An increase in the value of **N.Cdrx.Sleep.Dur.Total** may increase the probability of false SR detection in DRX sleep time, thereby increasing the uplink RBLER. Uplink RBLER = $(N.UL.SCH.HalfPiBPSK.ErrTB.Rbler + N.UL.SCH.QPSK.ErrTB.Rbler + N.UL.SCH.16QAM.ErrTB.Rbler + N.UL.SCH.64QAM.ErrTB.Rbler + N.UL.SCH.256QAM.ErrTB.Rbler) / (N.UL.SCH.HalfPiBPSK.TB + N.UL.SCH.QPSK.TB + N.UL.SCH.16QAM.TB + N.UL.SCH.64QAM.TB + N.UL.SCH.256QAM.TB) \times 100\%$

If the gNodeB configures SRs and CSI to be reported in the same timeslot and some UEs simultaneously report SRs and CSI in DRX sleep times, the gNodeB cannot correctly detect the SRs reported by these UEs. As a result, uplink data cannot be scheduled in a timely manner, increasing the uplink data transmission delay. To solve this problem, the following functions are introduced:

- Simultaneous reporting of SRs and CSI in DRX sleep times, which is controlled by the **DRX_SR_CSI_REPORT_SW** option of the **NRDUCellUePwrSaving.NrDuCellDrxAlgoSwitch** parameter. After this function is enabled, the gNodeB will correctly detect SRs reported by a UE when the UE simultaneously reports SRs and CSI in the DRX sleep time. For UEs that support simultaneous reporting of SRs and CSI in DRX sleep times, the uplink data transmission delay decreases. For UEs that do not support

simultaneous reporting of SRs and CSI in DRX sleep times, the uplink data transmission delay increases.

- Non-overlapping between CSI and SR reporting slots, which is controlled by the **DRX_CSI_SLOT_SHIFT_FOR_SR_SW** option of the **NRDUCellUePwrSaving.NrDuCellDrxAlgoSwitch** parameter. After this function is enabled, the gNodeB will not configure a UE's CSI reporting slot in which the UE reports an SR. This prevents the increase of uplink data transmission delay caused by overlapping CSI and SR reporting slots. Non-overlapping between CSI and SR reporting slots is supported only in low-frequency bands.

When configuring or reconfiguring DRX parameters (DRX-Config) for a UE using RRC signaling messages, the gNodeB cannot accurately determine the time when the DRX parameters take effect for the UE. There is a possibility that the DRX parameters have taken effect on the gNodeB side but still not on the UE side. In this scenario, the gNodeB incorrectly considers that the UE is in the DRX sleep time and stops scheduling for the UE. As a result, the UE access delay increases and the user-perceived throughput decreases. To address this issue, DRX configuration procedure optimization is introduced. This function is controlled by the **DRX_CONFIG_OPT_SW** option of the **NRDUCellUePwrSaving.NrDuCellDrxAlgoSwitch** parameter. After the DRX configuration procedure optimization function is enabled, the gNodeB optimizes the DRX parameter configuration and reconfiguration procedures. Specifically, new DRX configurations take effect on the gNodeB side only after the gNodeB receives a configuration response message from the UE, and the gNodeB does not stop scheduling the UE during the DRX configuration procedure, thereby increasing the user-perceived throughput.

If downlink enhancement for power saving is enabled for low-frequency TDD cells served by the MetaAAU (by selecting the **PWR_SAVING_DOWNLINK_ENH_SW** option of the **NRDUCellPdsch.DLLinkAdaptEnhancementSw** parameter), the network impacts are as follows:

- The increase in RRC connection reconfiguration messages may increase the service drop rate, CPU usage, and the proportion of MCS index 0, and decrease the MCS index for downlink PDSCH signaling scheduling and the average MCS index for downlink scheduling. The involved counters are as follows:
 - Average number of RBs used by signaling on the PDSCH (**N.PR.B.DL.SrbUsed.Avg**)
 - Average user-plane CPU usage of a gNodeB board (**VS.NRBoard.UPlane.CPULoad.Avg**)
 - Maximum user-plane CPU usage of a gNodeB board (**VS.NRBoard.UPlane.CPULoad.Max**)
 - Average Board CPU Usage (**VS.BBUBoard.CPULoad.Mean**)
 - Maximum Board CPU Usage (**VS.BBUBoard.CPULoad.Max**)
 - Service drop rates (**N.NsaDc.SgNB.AbnormRel**, **N.NsaDc.SgNB.Rel**, **N.NsaDc.SgNB.Rel.SgNBTrigger**)
 - Average MCS index for downlink scheduling ($(\text{N.ChMeas.PDSCH.MCS.0} + 1 \times \text{N.ChMeas.PDSCH.MCS.1} + 2 \times \text{N.ChMeas.PDSCH.MCS.2} + 3 \times \text{N.ChMeas.PDSCH.MCS.3} + \dots + 26 \times \text{N.ChMeas.PDSCH.MCS.26} + 27 \times \text{N.ChMeas.PDSCH.MCS.27} + 28 \times \text{N.ChMeas.PDSCH.MCS.28}) /$

$(N.ChMeas.PDSCH.MCS.0 + N.ChMeas.PDSCH.MCS.1 + N.ChMeas.PDSCH.MCS.2 + N.ChMeas.PDSCH.MCS.3 + \dots + N.ChMeas.PDSCH.MCS.26 + N.ChMeas.PDSCH.MCS.27 + N.ChMeas.PDSCH.MCS.28))$

- The proportion of DRX sleep time of RRC_CONNECTED UEs fluctuates.
 - If the DRX algorithm switch is turned on (by selecting the **BASIC_DRX_SW** option of the **NRDUCellUePwrSaving.NrDuCellDrxAlgoSwitch** parameter) but the BWP power saving switch is turned off (by deselecting the **BWP2_SWITCH** option of the **NRDUCellUePwrSaving.BwpPwrSavingSw** parameter), DRX takes effect on UEs that initially access the network after a delay because downlink enhancement for power saving takes effect. As a result, the proportion of UEs in the DRX sleep state may decrease.
 - If both the DRX algorithm switch (**BASIC_DRX_SW** option of the **NRDUCellUePwrSaving.NrDuCellDrxAlgoSwitch** parameter) and the BWP power saving switch (**BWP2_SWITCH** option of the **NRDUCellUePwrSaving.BwpPwrSavingSw** parameter) are selected:
 - DRX takes effect on UEs that initially access the network after a delay because downlink enhancement for power saving takes effect. As a result, the proportion of UEs in the DRX sleep state may decrease.
 - The duration for some UEs to work on BWP2 decreases and the duration for them to work on BWP1 increases because downlink enhancement for power saving takes effect. As a result, the service data transmission speed increases and the proportion of UEs in the DRX sleep state may increase.
 - If cell-level beam sensing is in centralized mode (by setting **NRDUCellBeamAlgo.BeamPerceiveMode** to **CENTRALIZED_MODE**) and downlink enhancement for power saving is enabled, the proportion of UEs in the DRX sleep state may decrease more significantly than when cell-level beam sensing is in distributed mode (by setting **NRDUCellBeamAlgo.BeamPerceiveMode** to **DISTRIBUTED_MODE**) and downlink enhancement for power saving is enabled.

Coexistence of DRX and gap-assisted measurement (controlled by the **DRX_AND_GAP_MEAS_COEXIST_SW** option of the **NRDUCellUePwrSaving.NrDuCellDrxAlgoSwitch** parameter) in SA networking may have the following impacts on the network:

- The service delay $(N.QoS.DL.PktDelayAirInterface.Time / N.QoS.DL.PktDelayAirInterface.Num + N.QoS.DL.PktDelaygNBDU.Time / N.QoS.DL.PktDelaygNBDU.Num)$ increases.
- The average uplink UE throughput (**User Uplink Average Throughput (DU)**) and average downlink UE throughput (**User Downlink Average Throughput (DU)**) decrease.

After exit of weak-coverage UEs from DRX is enabled (by selecting the **WEAK_COV_DRX_EXIT_SW** option of the **NRDUCellUePwrSaving.NrDuCellDrxAlgoSwitch** parameter) and some UEs in weak coverage areas exit DRX mode, the number of RLC retransmissions will reach

the maximum more quickly. As a result, these UEs are abnormally released, increasing the service drop rate.

The inactivity timer length adaptation function (enabled by setting the **NRDUCellUePwrSaving.IAAadaptiveAlgoSwitch** parameter to **ON_IPRAN_CA**) has the following impacts on the network in inter-gNdoeB CA scenarios:

- After the inactivity timer length adaptation function is enabled, the **N.ChMeas.CA.SCell.PDSCH** counter value increases if the traffic volume is sufficient.
- The inactivity timer length adaptation function prolongs the DRX active time and decreases the proportion of DRX sleep time. As a result, the power consumption of UEs with this function in effect increases.

When both "DRX status evaluation based on scheduling feedback" and PDCCH beam robustness are enabled, the following network impacts may occur:

- The average downlink UE throughput in a cell (**User Downlink Average Throughput (DU)**) and average uplink UE throughput in a cell (**User Uplink Average Throughput (DU)**) increase.
- The uplink and downlink RBLERs decrease.
 - Downlink RBLER = $(N.DL.SCH.QPSK.ErrTB.Rbler + N.DL.SCH.16QAM.ErrTB.Rbler + N.DL.SCH.64QAM.ErrTB.Rbler + N.DL.SCH.256QAM.ErrTB.Rbler) / (N.DL.SCH.QPSK.TB + N.DL.SCH.16QAM.TB + N.DL.SCH.64QAM.TB + N.DL.SCH.256QAM.TB) \times 100\%$
 - Uplink RBLER = $(N.UL.SCH.HalfPiBPSK.ErrTB.Rbler + N.UL.SCH.QPSK.ErrTB.Rbler + N.UL.SCH.16QAM.ErrTB.Rbler + N.UL.SCH.64QAM.ErrTB.Rbler + N.UL.SCH.256QAM.ErrTB.Rbler) / (N.UL.SCH.HalfPiBPSK.TB + N.UL.SCH.QPSK.TB + N.UL.SCH.16QAM.TB + N.UL.SCH.64QAM.TB + N.UL.SCH.256QAM.TB) \times 100\%$
- The downlink and uplink IBLERs increase.
 - Downlink IBLER = $(N.DL.SCH.QPSK.ErrTB.Ibler + N.DL.SCH.16QAM.ErrTB.Ibler + N.DL.SCH.64QAM.ErrTB.Ibler + N.DL.SCH.256QAM.ErrTB.Ibler) / (N.DL.SCH.QPSK.TB + N.DL.SCH.16QAM.TB + N.DL.SCH.64QAM.TB + N.DL.SCH.256QAM.TB) \times 100\%$
 - Uplink IBLER = $(N.UL.SCH.HalfPiBPSK.ErrTB.Ibler + N.UL.SCH.QPSK.ErrTB.Ibler + N.UL.SCH.16QAM.ErrTB.Ibler + N.UL.SCH.64QAM.ErrTB.Ibler + N.UL.SCH.256QAM.ErrTB.Ibler) / (N.UL.SCH.HalfPiBPSK.TB + N.UL.SCH.QPSK.TB + N.UL.SCH.16QAM.TB + N.UL.SCH.64QAM.TB + N.UL.SCH.256QAM.TB) \times 100\%$
- The average downlink and uplink CCE aggregation levels decrease.
 - Average uplink CCE aggregation level = $(1 \times N.CCE.UL.AggLvl1Num + 2 \times N.CCE.UL.AggLvl2Num + 4 \times N.CCE.UL.AggLvl4Num + 8 \times N.CCE.UL.AggLvl8Num + 16 \times N.CCE.UL.AggLvl16Num) / (N.CCE.UL.AggLvl1Num + N.CCE.UL.AggLvl2Num + N.CCE.UL.AggLvl4Num + N.CCE.UL.AggLvl8Num + N.CCE.UL.AggLvl16Num)$
 - Average downlink CCE aggregation level = $(1 \times N.CCE.DL.AggLvl1Num + 2 \times N.CCE.DL.AggLvl2Num + 4 \times N.CCE.DL.AggLvl4Num + 8 \times N.CCE.DL.AggLvl8Num + 16 \times N.CCE.DL.AggLvl16Num) /$

$$(N.CCE.DL.AggLvl1Num + N.CCE.DL.AggLvl2Num + N.CCE.DL.AggLvl4Num + N.CCE.DL.AggLvl8Num + N.CCE.DL.AggLvl16Num)$$

- The DTX proportion decreases. In this case, the following are true: (1) When the number of UEs is small, the CCE aggregation level decreases and the values of the following indicators decrease. (2) When the number of UEs is large, the number of UEs in DRX active time increases, the probability of scheduling resource conflicts increases, and the values of the following indicators increase.
 - Uplink CCE allocation failure rate = $(N.CCE.UL.AllocReq.Num - N.CCE.UL.AggLvl1Num - N.CCE.UL.AggLvl2Num - N.CCE.UL.AggLvl4Num - N.CCE.UL.AggLvl8Num - N.CCE.UL.AggLvl16Num) / N.CCE.UL.AllocReq.Num$
 - Downlink CCE allocation failure rate = $(N.CCE.DL.AllocReq.Num - N.CCE.DL.AggLvl1Num - N.CCE.DL.AggLvl2Num - N.CCE.DL.AggLvl4Num - N.CCE.DL.AggLvl8Num - N.CCE.DL.AggLvl16Num) / N.CCE.DL.AllocReq.Num$
- The SgNB-triggered abnormal SgNB release rate, indicated by $(N.NsaDc.SgNB.AbnormRel / N.NsaDc.SgNB.Rel) \times 100\%$, increases.
- The QoS flow service drop rate, indicated by $(N.QosFlow.AbnormRel / (N.QosFlow.AbnormRel + N.QosFlow.NormRel)) \times 100\%$, increases.

4.3 Requirements

4.3.1 Licenses

None

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

None

4.3.2.2 Mutually Exclusive Functions

None

4.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intra-RAT ANR	NR_NR_ANR_SW option of the NRCellAlgoSwitch . <i>AnrSwitch</i> parameter	<i>ANR</i>	• The UE can perform reportCGI measurements only after entering the DRX sleep time. • When the gNodeB triggers a UE to read the NCGIs, the UE automatically uses a long DRX cycle. The setting of the BASIC_DRX_SW option of the NRDUCellUePwrSaving.<i>NrDuCellDrxAlgoSwitch</i> parameter does not affect the configuration of the long DRX cycle initiated by intra-RAT ANR.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Inter-RAT ANR	NR_EUTRAN_ANR_SW option of the NRCellAlgoSwitch . <i>AnrSwitch</i> parameter	<i>ANR</i>	• The UE can perform reportCGI measurements only after entering the DRX sleep time. • When the gNodeB triggers a UE to read the ECGIs, the UE automatically uses a long DRX cycle. The setting of the BASIC_DRX_SW option of the NRDUCellUePwrSaving.<i>NrDuCellDrxAlgoSwitch</i> parameter does not affect the configuration of the long DRX cycle initiated by inter-RAT ANR.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	SRS resource management	None	<i>Channel Management</i>	- For low-frequency cells: - Periodic SRS: The gNodeB does not receive or measure SRS from UEs in DRX sleep time. Therefore, the quantity of valid SRSs received by the gNodeB decreases. - Aperiodic SRS: The gNodeB does not schedule SRSs from UEs in DRX sleep time. Therefore, the quantity of valid SRSs received by the gNodeB decreases. - For high-frequency cells: Aperiodic SRS: The base station performs scheduling only within the length of the Inactivity Timer triggered by common data transmission or the On Duration Timer. Therefore, the quantity of valid SRSs received by the gNodeB decreases and the

RAT	Function Name	Function Switch	Reference	Description
				measurement interval increases. For details about periodic and aperiodic SRSs, see <i>Channel Management</i> .

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	CSI-RS resource management	None	<i>Channel Management</i>	• For low-frequency cells: – Periodic CSI-RS for CM: The gNodeB does not check CSI-RS measurement reports from UEs in DRX sleep times. As a result, the number of valid CSI-RS measurement reports received by the gNodeB decreases. – Aperiodic CSI-RS for CM: UEs send CSI-RS measurement reports on the PUSCH, which may cause continuous scheduling of uplink data for UEs. As a result, the UEs cannot enter DRX sleep times. • For high-frequency cells: – Periodic CSI-RS for BM: The gNodeB does not check CSI-RS measurement reports from UEs in DRX sleep time. As a result, the number of

RAT	Function Name	Function Switch	Reference	Description
				valid CSI-RS measurement reports received by the gNodeB decreases. - Aperiodic CSI-RS for BM: The gNodeB schedules the aperiodic CSI measurement results reported by UEs only within the length of the Inactivity Timer triggered by common data transmission or the On Duration Timer. As a result, the number of valid CSI reports received by the gNodeB decreases and the reporting interval increases. For details about periodic and aperiodic CSI-RS, see <i>Channel Management</i>.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	Uplink preallocation	NRDUCellPusch.UlPreallocationSwitch	<i>Uplink Scheduling</i>	- After the uplink basic preallocation function (controlled by the UL_BASIC_PREALLOCATION_SWITCH option of the NRDUCellPusch.UlPreallocationSwitch parameter) is enabled, UEs cannot enter the sleep time. - After uplink smart preallocation function (corresponding to the UL_SMART_PREALLOCATION_SWITCH option of the NRDUCellPusch.UlPreallocationSwitch parameter) is enabled, UEs cannot enter the sleep time within the duration for uplink smart preallocation. - After uplink preallocation for sparse-packet services is enabled (by selecting the SPARSE_DIFF_PROC_SW option of the gNodeBParam.ServiceDiffProcSwitch parameter and the SPARSE_DIFF_SCH_SW option of the NRDUCellAlgoS

RAT	Function Name	Function Switch	Reference	Description
				<i>witch.ServiceDiffSwitch</i> parameter), UEs performing sparse-packet services cannot enter the sleep time.
Low-frequency TDD	Basic VoNR Functions	VONR_SW option of the NRCCellAlgoS witch.VonrS witch parameter	<i>VoNR</i>	When the gNodeB RLC parameter group is associated with 5QI 1 bearers: • If NRDUCellQciBearer.DuDrxParamGroupId or NRCCellQciBearer.DrxParamGroupId is set to the default value 255, DRX does not take effect and VoNR services are not affected. • If neither NRDUCellQciBearer.DuDrxParamGroupId nor NRCCellQciBearer.DrxParamGroupId is set to the default value, DRX is not affected. However, the transmission delay of voice packets increases, affecting the voice quality of downlink VoNR services.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intra-base-station DL CoMP	INTRA_GNB_DL_JT_SW option of the NRDUCellAIgoSwitch.CompSwitch parameter	<i>CoMP</i>	- After the DRX function is enabled, UEs are less likely to be identified as intra-base-station joint transmission UEs. - After DRX is enabled: if the function of DRX exit of UEs performing full-buffer services (controlled by the FULL_BUFFER_UE_EXIT_DRX_SW option of the NRDUCellUePwrSaving.NrDuCellDrxAlgoSwitch parameter) is enabled, DRX does not take effect on UEs identified as intra-base-station joint transmission UEs; if the function of DRX exit of UEs performing full-buffer services is disabled, UEs identified as intra-base-station joint transmission UEs cannot enter the DRX sleep time because the aperiodic CSI-RS for CM is used.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Power saving BWP (low-frequency TDD)	BWP2_SWITCH option of the NRDUCellUePwrSaving.BwpPwrSavingSw parameter	<i>UE Power Saving</i>	- All UEs working in BWP2 aperiodically report CSI-RS for CM measurement results if both power saving BWP and DRX are enabled, NRDUCellPucch.CsiDedicatedRbNum is set to RB2, and the PUCCH_RBRES_ADAPTIVE_SWITCH option of the NRDUCellPucch.PucchAlgoSwitch parameter is deselected. Because such UEs aperiodically report CSI-RS for CM measurement results on the PUSCH, uplink data of these UEs may be scheduled continuously. As a result, these UEs cannot enter the DRX sleep time. - If both power saving BWP and DRX are enabled and the gNBDUDrxParamGroup.DrxInactivityTimer parameter is set to a value less than 20 ms, UEs may stay in BWP2 and cannot switch to BWP1. - If both power saving BWP and DRX are enabled

RAT	Function Name	Function Switch	Reference	Description
				and the BWP2_APER_CSI_UE_DRX_ENH_SW option of the NRDUCellUePwrSaving.NrDuCellDrxAlgoSwitch parameter is selected, UEs in BWP2 do not aperiodically report CSI-RS for CM measurement results even though configured to do so when there is no uplink data transmission in non-On Duration periods. This way, these UEs can enter the DRX sleep time and the number of CSI-RS for CM measurement reports decreases, affecting the network performance. For details about the impact analysis, see 4.2.2 Impacts. • If both power saving BWP and DRX are enabled and the gNodeB triggers a switchover from BWP2 to BWP1 during the DRX sleep time, the DCI for switching from BWP2 to BWP1 may fail to be delivered. As a result, the

RAT	Function Name	Function Switch	Reference	Description
				gNodeB stops scheduling downlink data transmission for the UE during the BWP switchover, leading to downlink scheduling failures. To address this issue, the DRX_BWP2_OPT_SW option of the NRDUCellUePwrSaving.BwpPwrSavingSw parameter can be selected so that the gNodeB triggers UE switching from BWP2 to BWP1 only when the DRX inactivity timer is running. • When the BWP2_PERIOD_CSI_SW option of the NRDuCellUePwrSaving.BwpPwrSavingSw parameter is selected, periodic PUCCH CSI reporting resources can be allocated to BWP2 UEs. This helps increase the proportion of UEs in the DRX sleep time and improve power saving gains. However, the average uplink and downlink UE

RAT	Function Name	Function Switch	Reference	Description
				throughput decreases, the delay increases, and the measured CQI value may increase or decrease. • After downlink enhancement for power saving is enabled, the duration in which UEs stay in the narrow-bandwidth BWP2 decreases, accelerating data transmission and possibly increasing the proportion of UEs in the DRX sleep time. As a result, the number of false SR detections, BLER, number of retransmissions, and number of used CCEs and PRBs increase. For details about the impact of DRX sleep time on network performance, see 4.2.2 Impacts.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intra-band CA	INTRA_BAND_CA_SW option of the NRDUCellAIgoSwitch.CaAlgoSwitch parameter INTRA_BAND_UL_CA_SW option of the NRDUCellAIgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	Data split for a UE is affected if the UE's DRX state is inconsistent between its PCell and SCells. When intra-band CA and DRX are both enabled, it is recommended that collaboration between CA and DRX be enabled. The collaboration is controlled by the CA_AND_DRX_STATUS_COLLAB_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch parameter.
Low-frequency TDD	Intra-FR inter-band CA	INTRA_FR_INTER_BAND_CA_SW option of the NRDUCellAIgoSwitch.CaAlgoSwitch parameter INTRA_FR_INTER_BAND_UL_CA_SW option of the NRDUCellAIgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	Data split for a UE is affected if the UE's DRX state is inconsistent between its PCell and SCells. When intra-FR inter-band CA and DRX are both enabled, it is recommended that collaboration between CA and DRX be enabled. The collaboration is controlled by the CA_AND_DRX_STATUS_COLLAB_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch parameter.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink data transmission path selection	LTE: NSA_DC_UL_PATH_SELECTION_SW option of the NsaDcMgmtConfig.NsaDcAlgoSwitch parameter NR: NSA_DC_UL_PATH_SELECTION_SW option of the NRCellNsaDcConfig.NsaDcAlgoSwitch parameter	<i>EPC-based NSA Performance Enhancement</i>	After DRX is enabled, if DRX exit of UEs performing full-buffer services is enabled (by selecting the FULL_BUFFER_UE_EXIT_DRX_SW option of the NRDUCellUePwrSaving.NrDuCellDrxAlgoSwitch parameter) and the uplink data transmission path selection function takes effect for a UE, the UE is a full buffer UE. In this case, DRX does not take effect.
Low-frequency TDD	QCI-specific scheduling based on the downlink target IBLER	DL_IBLER_QCI_SW of the gNBDUSchParamGroup.ScheduleOptSwitch parameter	<i>Service-differentiated Experience Guarantee</i>	It is not recommended that DRX be enabled for bearers carrying differentiated low-latency services.
Low-frequency TDD High-frequency TDD	Low latency and high reliability	HIGH_RELIABILITY_BASIC_SW option of the NRDUCellAlgoSwitch.HighReliabilitySwitch parameter	<i>URLLC</i>	It is not recommended that DRX be enabled for bearers carrying low-latency and high-reliability services. In high-frequency bands, DRX does not take effect on UEs running low-latency and high-reliability services.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag	<i>High Speed Mobility</i>	To enable DRX in high-speed cells, the differentiated DRX function must be enabled (by selecting the LOW_SPEED_UE_DRX_SW option of the NRCellAlgoSwitch.HighSpeedDedPolicySw parameter). In high-speed cells, DRX takes effect only on low-speed UEs.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Co-MM	INTRA_GNB_SU_COMM_SW option of the NRDUCellCoIlabServ.MultiCellMimoSwitch parameter	<i>Co-MM (Low-Frequency TDD)</i>	- After DRX is enabled (by selecting the BASIC_DRX_SW option of the NRDUCellUePwrSaving.NrDuCellDrxAlgoSwitch parameter), the probability that UEs are identified as Co-MM UEs decreases. - After DRX is enabled, if exit of UEs performing full-buffer services from DRX (controlled by the FULL_BUFFER_U_EXIT_DRX_SW option of the NRDUCellUePwrSaving.NrDuCellDrxAlgoSwitch parameter) is enabled, DRX (sleep or active time) does not take effect for Co-MM UEs. If exit of UEs performing full-buffer services from DRX is disabled, Co-MM UEs cannot enter the DRX sleep time because the aperiodic CSI-RS for CM is used.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	QCI-based uplink basic preallocation	UL_PREALLOCATION_QCI_SW option of the gNBDUSchParamGroup.ScheduleOptionSwitch parameter	<i>Service-differentiated Experience Guarantee</i>	When both DRX (controlled by the BASIC_DRX_SW option of the NRDUCellUePwrSaving.NrDuCellDrxAlgorithmSwitch parameter) and QCI-based uplink basic preallocation are enabled: • If there is no voice service, the actual uplink preallocation interval may be less than the configured value of the gNBDUSchParamGroup.UlPreallocationInterval parameter. • If there are voice services, the actual uplink preallocation interval is the same as the configured value of the gNBDUSchParamGroup.UlPreallocationInterval parameter.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	QCI-based uplink smart preallocation	UL_SMART_PREALLOC_QCI_SW option of the gNBDUSchParamGroup.ScheduleOptionSwitch parameter	<i>Service-differentiated Experience Guarantee</i>	When both DRX (controlled by the BASIC_DRX_SW option of the NRDUCellUePwrSaving.NrDuCellDrxAlgorithmSwitch parameter) and QCI-based uplink smart preallocation are enabled: • If there is no voice service, the actual uplink preallocation interval may be less than the configured value of the gNBDUSchParamGroup.UlPreallocationInterval parameter. • If there are voice services, the actual uplink preallocation interval is the same as the configured value of the gNBDUSchParamGroup.UlPreallocationInterval parameter.
Low-frequency TDD	UE-characteristic-based SRS period adaptation	USER_CHARACTER_SRS_ADAPT_SW option of the NRDUCellSrs.SrsAlgorithmSwitch parameter	<i>Channel Management</i>	If UE-characteristic-based SRS period adaptation is enabled, the DRX sleep time will be shortened to improve the SRS-based weight performance.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	Inter-FR CA	INTER_FR_F R1TOFR2_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	DRX can take effect on an inter-FR CA UE only when DRX is enabled for all the serving cells of the UE. The DRX settings for the low-frequency PCell apply to the UE.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Proactive avoidance of remote interference	SRS_RIM_INTRF_AVOID_SW and SRS_INTRF_COORD_S_SLOT_SW options of the NRDUCellSrs.SrsAlgoExtSwitch parameter	<i>Remote Interference Management (Low-Frequency TDD)</i>	When both active remote interference avoidance (controlled by the SRS_RIM_INTRF_AVOID_SW and SRS_INTRF_COORD_S_SLOT_SW options of the NRDUCellSrs.SrsAlgoExtSwitch parameter) and DRX (controlled by the BASIC_DRX_SW option of the NRDUCellUePwrSaving.NrDuCellDrxAlgoSwitch parameter) are enabled, the DRX On Duration may be prolonged or shortened. • After SRS resources are migrated to uplink slots, the SRS time-domain position as well as the number and time-domain positions of PUCCH resources change. As a result, the On Duration may change because it is related to the PUCCH/SRS time-domain positions. • After SRS resources are migrated to uplink slots, the interference to SRSs decreases and the downlink performance improves. Given

RAT	Function Name	Function Switch	Reference	Description
				the same amount of data, it takes shorter time for a UE to receive data and the UE is more likely to enter the sleep time. Therefore, the On Duration may be shortened.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	SRS remote interference avoidance	SRS_RIM_IN TRF_AVOID_SW option of the NRDUCellSrsSrsAlgoExtSwitch parameter	None	- When SRS remote interference avoidance and DRX are both enabled, the DRX On Duration may be prolonged or shortened. - After SRS resources are migrated to uplink slots, the SRS time-domain position as well as the number and time-domain positions of PUCCH resources change. As a result, the On Duration may change because it is related to the PUCCH/SRS time-domain positions. - After SRS resources are migrated to uplink slots, the interference to SRSs decreases and the downlink performance improves. Given the same amount of data, it takes shorter time for a UE to receive data and the UE is more likely to enter the sleep time. Therefore, the On Duration may be shortened.

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	mmWave beam management	None	<i>mmWave Beam Management (High-Frequency TDD)</i>	After DRX takes effect, the interval between SSB and CSI beam sweeping reporting increases. As a result, the number of valid beam sweeping measurement reports received by the gNodeB from UEs decreases, and the timeliness of UE beam tracking decreases. In addition, when narrow beams are used for TRS, delayed beam tracking may cause inaccurate downlink time-frequency offset measurement.
High-frequency TDD	Uplink timing	None	<i>5G Networking and Signaling</i>	The gNodeB does not schedule SRSs in DRX sleep time. Therefore, the gNodeB will receive fewer valid SRSs from UEs, which may cause inaccurate uplink timing and increase the uplink BLER.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Coordinated interference management	INTRA_GNB_DL_CS_CBF_SW and INTER_GNB_DL_CS_CBF_SW options of the NRDUCellAIgoSwitch.CompSwitch parameter	<i>Coordinated Interference Management (Low-Frequency TDD)</i>	- After DRX is enabled, the probability that CS/CBF takes effect decreases. - After DRX is enabled, it is recommended that the FULL_BUFFER_UE_EXIT_DRX_SW option of the NRDUCellUePwrSaving.NrDuCellDrxAlgoSwitch parameter be selected for both the serving and cooperating cells.
Low-frequency TDD	NR DU cell resource management between network slices	RB_DYNAMIC_CONTROL_SW option of the NRDUCellAIgoSwitch.NetworkSliceAIgoSwitch parameter	<i>Network Slicing</i>	When both DRX and NR DU cell resource management between network slices are enabled and the latter function is configured with a network slice group with the TTI-based control policy, all UEs served by the network slice group with the TTI-based control policy in the NR DU cell will not enter the DRX state. In this situation, DRX used for reportCGI measurements is not included.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	U-TDOA- and AoA-based high-precision positioning	AOA_UTDOA_HYBRID_LOCATION_SW option of the NRCellAlgoSwitch.LcsSwitch parameter	<i>Positioning</i>	DRX does not take effect on UEs with U-TDOA- and AoA-based high-precision positioning in effect.
Low-frequency TDD	Device-pipe synergy preallocation	UL_PREFERENTIAL_SCH_SW option of the NRDUCellQciBearer.ServiceDiffAlgoSwitch parameter	<i>FWA</i>	UEs with device-pipe synergy preallocation in effect may fail to enter the sleep time.
Low-frequency TDD	3GPP R15-compliant uplink slot aggregation	VONR_UL_SLOT_AGGREGATION_SW option of the NRDUCellULSch.VoiceULSchSwitch parameter	<i>VoNR</i>	If a UE with VoNR enabled is in the uplink slot aggregation state, the gNodeB no longer instructs it to enter DRX mode. If the UE has been in DRX mode before entering the uplink slot aggregation state, the gNodeB will instruct it to exit DRX mode. If the UE exits the uplink slot aggregation state, the gNodeB will deliver DRX-related configurations to it again.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	3GPP R16-compliant uplink slot aggregation	VOICE_R16_UL_SLOT_AGG_SW option of the NRDUCellUlschSwitch parameter	<i>VoNR</i>	If a UE with VoNR enabled is in the uplink slot aggregation state, the gNodeB no longer instructs it to enter DRX mode. If the UE has been in DRX mode before entering the uplink slot aggregation state, the gNodeB will instruct it to exit DRX mode. If the UE exits the uplink slot aggregation state, the gNodeB will deliver DRX-related configurations to it again.
Low-frequency TDD	SRS field strength-based positioning	SRS_RSRP_LOCATION_SW option of the NRCCellAlgoSwitch.LcsSwitch parameter	<i>Positioning</i>	DRX does not take effect on UEs with SRS field strength-based positioning in effect.
Low-frequency TDD	U-TDOA-based positioning/TOA fingerprint-based positioning	UTDOA_SW option of the NRCCellAlgoSwitch.LcsSwitch parameter	<i>Positioning</i>	DRX does not take effect on UEs with U-TDOA-based positioning/TOA fingerprint-based positioning in effect.
Low-frequency TDD	Inter-cell interference avoidance	INTER_CELL_INTRF_AVOID_SW option of the NRDUCellCollabServ.MultiCellMimoSwitch parameter	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	When DRX takes effect, inter-cell interference avoidance may not take effect due to failures to determine correlations. If this occurs, the benefits of inter-cell interference avoidance change.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Time-domain power saving BWP	TDM_BWP2_SWITCH option of the NRDUCellUePwrSaving.BwpPwrSavingSw parameter	<i>UE Power Saving</i>	When both time-domain power saving BWP and DRX are enabled: • If the duration configured using the gNBDUDrxParamGroup.DrxInactivityTimer parameter is less than 20 ms, UEs may remain working in BWP2 and cannot be switched to BWP1. • If the BWP2_APER_CSI_UE_DRX_ENH_SW option of the NRDUCellUePwrSaving.NrDuCellDrxAlgoSwitch parameter is selected, UEs in BWP2 do not aperiodically report CSI-RS for CM measurement results even though configured to do so when there is no uplink data transmission in non-On Duration periods. This way, these UEs can enter the DRX sleep time and the number of CSI-RS for CM measurement reports decreases, affecting network performance. For details about the

RAT	Function Name	Function Switch	Reference	Description
				impact analysis, see 4.2.2 Impacts. • If the base station triggers a BWP2-to-BWP1 switching during the DRX sleep time, the corresponding DCI may fail to be delivered. As a result, the base station stops downlink scheduling for the UE during the BWP switching, leading to downlink scheduling failures. After the DRX_BWP2_OPT_SW option of the NRDUCellUePwrSaving.BwpPwrSavingSw parameter is selected, the base station can trigger UE switching from BWP2 to BWP1 only when the DRX inactivity timer is running.

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

4.3.4 Cells

In high-frequency NR TDD scenarios, when slot configuration (specified by **NRDUCell.SlotAssignment**) is set to **2_3_DSUUU** for a cell, the DRX feature (controlled by the **BASIC_DRX_SW** option of the **NRDUCellUePwrSaving.NrDuCellDrxAlgoSwitch** parameter) cannot be enabled.

4.3.5 Others

To support DRX basic functions, UEs must support the "longDRX-Cycle" field in the MAC-ParametersXDD-Diff message. To support the short DRX cycle function, UEs must also support the "shortDRX-Cycle" field. For details about the fields, see section 6.3.3 UE capability information elements in 3GPP TS 38.331 V16.4.1.

4.4 Operation and Maintenance

4.4.1 Data Configuration

4.4.1.1 Data Preparation

Table 4-9 and **Table 4-10** describe the parameters used for function activation and optimization, respectively. This section does not describe parameters related to cell establishment.

QCI-specific DRX parameter settings require that bearers with specified QCIs have been set up for UEs. Therefore, before using this function, ensure that bears of relevant QCIs have been set up. For details, see *QoS Management*.

Table 4-9 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
NR DU Cell DRX Algorithm Switch	NRDUCellUEPwrSaving.NrDuCellDrxAlgorithmSwitch	BASIC_DRX_SW	Set this parameter based on the network plan.
DRX Parameter Group ID	NRCeIIQciBeareer.DrxParamGroupI d	None	For standardized QCI, this parameter is set by default and does not need to be manually configured. For extended QCI, this parameter needs to be configured based on the network plan. Before enabling DRX, ensure that all QCI used by UEs are bound to valid DRX parameter groups. The QCI added to the NRCeIIQciBeareer MO must exist in the gNBQciBeareer MO.
DU DRX Parameter Group ID	NRDUCeIIQciBeareer.DuDrxParamGroupI d	None	This parameter is set for standardized QCI by default and does not need to be manually configured. For extended QCI, this parameter needs to be configured based on the network plan. Before enabling DRX, ensure that all QCI used by UEs are bound to valid DRX parameter groups. The QCI added to the NRDUCeIIQciBeareer MO must exist in the gNBQciBeareer MO.

Table 4-10 Parameters used for optimization

Parameter Name	Parameter ID	Option	Setting Notes
DRX Parameter Group ID	gNBDrxParamGroup.DrxParamGroupI d	None	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Option	Setting Notes
Long Cycle	gNBDrxParamGroup.LongCycle	None	A larger parameter value results in a longer sleep time and longer delay of UEs. Set this parameter based on the network plan.
Short Cycle	gNBDrxParamGroup.ShortCycle	None	A larger parameter value results in a longer sleep time and longer delay of UEs. Set this parameter based on the network plan.
Short Cycle Timer	gNBDrxParamGroup.ShortCycleTimer	None	A larger parameter value results in a longer short DRX cycle and shorter delay of UEs. Set this parameter based on the network plan.
DU DRX Parameter Group ID	gNBDUDrxParamGroup.DuDrxParameterGroupID	None	Set this parameter based on the network plan.
On Duration Timer	gNBDUDrxParamGroup.OnDurationTimer	None	A larger parameter value results in a longer active time and shorter delay of UEs. Set this parameter based on the network plan.

Parameter Name	Parameter ID	Option	Setting Notes
DRX Inactivity Timer	gNBDUDrxParamGroup.DrxInactivityTimer	None	A larger parameter value results in a longer active time extension after a UE receives new data during the On Duration Timer. The length of this timer must be longer than or equal to the time difference between SR sending and scheduling. Otherwise, user experience is affected and service drops may occur. Therefore, it is recommended that this timer be set to 20 ms or longer.
DRX Retransmission Timer	gNBDUDrxParamGroup.DrxRetransTimer	None	This parameter specifies the length of the DL Retransmission Timer and UL Retransmission Timer. Set this parameter based on the network plan.
NR DU Cell DRX Algorithm Switch	NRDUCellUePwrSaving.NrDuCellDrxAlgoSwitch	BWP2_APER_CSI_UE_DRX_ENH_SW	Select this option based on the network plan.
NR DU Cell DRX Algorithm Switch	NRDUCellUePwrSaving.NrDuCellDrxAlgoSwitch	FULL_BUFFER_UE_EXIT_DRX_SW	Select this option.
NR DU Cell DRX Algorithm Switch	NRDUCellUePwrSaving.NrDuCellDrxAlgoSwitch	UL_BASED_DRX_STATUS_DETECT_SW	Select this option.

Parameter Name	Parameter ID	Option	Setting Notes
DRX Sleep State Detect Threshold	NRDUCellUePwrS aving.DrxSleepSta teDetectThld	None	Set this parameter based on the network plan. A smaller value of this parameter results in more timely identification of false SR detections, increasing the number of scheduled UEs and cell throughput in heavy-load cells. However, false SR detections may be incorrectly identified, decreasing the UE throughput. A larger value of this parameter results in a longer time for identifying false SR detections. This decreases the number of scheduled UEs and cell throughput on a heavily loaded network and increases the UE throughput on a lightly loaded network.
PDCCH Algorithm Switch	NRDUCellPdcch.Pd cchAlgoSwitch	PDCCH_BEAM_RO BUST_SW	Select this option if PDCCH beam robustness is required.
NR DU Cell DRX Algorithm Switch	NRDUCellUePwrS aving.NrDuCellDrx AlgoSwitch	DRX_CONFIG_OP T_SW	Select this option.

Parameter Name	Parameter ID	Option	Setting Notes
BWP Power Saving Switch	NRDUCellUePwrSaving.BwpPwrSavingSw	DRX_BWP2_OPT_SW	Select this option.
NR DU Cell DRX Algorithm Switch	NRDUCellUePwrSaving.NrDuCellDrxAlgoSwitch	DRX_CSI_SLOT_SHIFTS_FOR_SR_SW	Select this option based on the network plan.
DL Link Adaptation Enhancement Switch	NRDUCellPdsch.DLLinkAdaptEnhancementSw	PWR_SAVING_DOWNLINK_ENH_SW	Select this option.
NR DU Cell DRX Algorithm Switch	NRDUCellUePwrSaving.NrDuCellDrxAlgoSwitch	DRX_AND_GAP_MEAS_COEXIST_SW	Select this option based on the network plan.
NR DU Cell DRX Algorithm Switch	NRDUCellUePwrSaving.NrDuCellDrxAlgoSwitch	WEAK_COV_DRX_EXIT_SW	Select this option if enabling exit of weak-coverage UEs from DRX is required.
Weak Coverage Threshold Offset	NRDUCellUePwrSaving.WeakCoverageThldOffset	None	Configure this parameter if enabling exit of weak-coverage UEs from DRX is required. Set this parameter to its recommended value.
Initial DMRS RSRP Weak Coverage Threshold	NRDUCellDmrs.InitialDmrsRsrpWeakCoverageThld	None	Configure this parameter if enabling exit of weak-coverage UEs from DRX is required. Set this parameter to its recommended value.

Parameter Name	Parameter ID	Option	Setting Notes
Initial DMRS SINR Weak Coverage Threshold	NRDUCellDmrs.InitialDmrsSinrWeakCoverageThld	None	Configure this parameter if enabling exit of weak-coverage UEs from DRX is required. Set this parameter to its recommended value.
Weak Coverage CQI Threshold	NRDUCellUePwrSaving.WeakCoverageCqiThld	None	Configure this parameter if enabling exit of weak-coverage UEs from DRX is required. Set this parameter to its recommended value.
Weak Coverage UL SINR Threshold	NRDUCellUePwrSaving.WeakCoverageULSinrThld	None	Configure this parameter if enabling exit of weak-coverage UEs from DRX is required. Set this parameter to its recommended value.
InactiveTimer Adaptive Algorithm Switch	NRDUCellUePwrSaving.IAAdaptiveAlgorithmSwitch	None	In inter-gNodeB CA scenarios: - To improve gains offered by inter-gNodeB CA, set this parameter to ON_IPRAN_CA. - To improve UE power saving performance, set this parameter to OFF.

4.4.1.2 Using MML Commands

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

For high-frequency TDD cells

```
//Enabling the DRX function for a cell
MOD NRDUCELLUEPWRSAVING: NrDuCellId=0, NrDuCellDrxAlgoSwitch=BASIC_DRX_SW-1;
//Adding or binding DRX parameter groups is not required for QCI 1, QCI 65, QCI 66, and QCI 69 because
DRX is not recommended for them. The system adds and binds DRX parameter groups by default for QCI 2,
QCI 3, QCI 5, QCI 7, QCI 75, QCI 79, QCI 4, QCI 6, QCI 8, QCI 9, and QCI 70. For QCIs other than the
preceding ones, you must manually add and bind the DRX parameter groups as follows to use the DRX
function.
//In high-frequency scenarios, you are advised to add only a long DRX cycle. The following is an example of
adding a DRX parameter group:
ADD GNBDRXPARAMGROUP: DrxParamGroupId=2, LongCycle=MS320;
ADD GNBUDRXPARAMGROUP: DuDrxParamGroupId=2, OnDurationTimer=MS50,
DrxInactivityTimer=MS100, DrxRetransTimer=SL16;
//Binding the DRX parameter group to the QCI (QCI 100 is used as an example)
MOD NRCELLQCIBEARER: NrCellId=0, Qci=100, DrxParamGroupId=2;
MOD NRDUCELLQCIBEARER: NrDuCellId=0, Qci=100, DuDrxParamGroupId=2;
```

For low-frequency TDD cells

```
//Enabling the DRX function for a cell
MOD NRDUCELLUEPWRSAVING: NrDuCellId=0, NrDuCellDrxAlgoSwitch=BASIC_DRX_SW-1;
//Adding or binding DRX parameter groups is not required for QCI 1, QCI 65, QCI 66, and QCI 69 because
DRX is not recommended for them. The system adds and binds DRX parameter groups by default for QCI 2,
QCI 3, QCI 5, QCI 7, QCI 75, QCI 79, QCI 4, QCI 6, QCI 8, QCI 9, and QCI 70. For QCIs other than the
preceding ones, you must manually add and bind the DRX parameter groups as follows to use the DRX
function.
//Adding a DRX parameter group for only a long DRX cycle in low-frequency scenarios
ADD GNBDRXPARAMGROUP: DrxParamGroupId=2, LongCycle=MS320;
ADD GNBUDRXPARAMGROUP: DuDrxParamGroupId=2, OnDurationTimer=MS10,
DrxInactivityTimer=MS100, DrxRetransTimer=SL16;
//Adding DRX parameter groups for long and short DRX cycles in low-frequency scenarios
ADD GNBDRXPARAMGROUP: DrxParamGroupId=2, LongCycle=MS320, ShortCycle=MS80,
ShortCycleTimer=1;
ADD GNBUDRXPARAMGROUP: DuDrxParamGroupId=2, OnDurationTimer=MS10,
DrxInactivityTimer=MS100, DrxRetransTimer=SL16;
//Binding the DRX parameter group to the QCI (QCI 100 is used as an example)
MOD NRCELLQCIBEARER: NrCellId=0, Qci=100, DrxParamGroupId=2;
MOD NRDUCELLQCIBEARER: NrDuCellId=0, Qci=100, DuDrxParamGroupId=2;
```

Optimization Command Examples

For high-frequency TDD cells

```
//Enabling DRX exit of UEs performing full-buffer services
MOD NRDUCELLUEPWRSAVING: NrDuCellId=0, NrDuCellDrxAlgoSwitch=FULL_BUFFER_UE_EXIT_DRX_SW-1;
```

```
//Enabling DRX status evaluation based on scheduling feedback over the PUSCH
MOD NRDUCELLUEPWRSAVING: NrDuCellId=0,
NrDuCellDrxAlgoSwitch=UL_BASED_DRX_STATUS_DETECT_SW-1;
//Configuring the DRX sleep time detection threshold
MOD NRDUCELLUEPWRSAVING: NrDuCellId=0, DrxSleepStateDetectThld=2;
//Enabling DRX configuration procedure optimization
MOD NRDUCELLUEPWRSAVING: NrDuCellId=0, NrDuCellDrxAlgoSwitch=DRX_CONFIG_OPT_SW-1;
```

For low-frequency TDD cells

```
//Modifying the DRX parameter settings for a long DRX cycle
MOD GNBDRXPARAMGROUP: DrxParamGroupId=2, LongCycle=MS40;
MOD GNBUDRXPARAMGROUP: DuDrxParamGroupId=2, OnDurationTimer=MS10,
DrxInactivityTimer=MS20, DrxRetransTimer=SL16;
//Modifying the DRX parameter settings for long and short DRX cycles
MOD GNBDRXPARAMGROUP: DrxParamGroupId=2, LongCycle=MS40, ShortCycle=MS20, ShortCycleTimer=1;
MOD GNBUDRXPARAMGROUP: DuDrxParamGroupId=2, OnDurationTimer=MS10,
DrxInactivityTimer=MS20, DrxRetransTimer=SL16;
//Modifying the binding relationship between a DRX parameter group and a QCI
MOD NRCELLQCIBEARER: NrCellId=0, Qci=100, DrxParamGroupId=0;
MOD NRDUCELLQCIBEARER: NrDuCellId=0, Qci=100, DuDrxParamGroupId=0;
//Enabling coordination between the DRX feature and UEs adopting aperiodic CSI-RS for CM reporting on
BWP2 in low-frequency scenarios
MOD NRDUCELLUEPWRSAVING: NrDuCellId=0,
NrDuCellDrxAlgoSwitch=BWP2_APER_CSI_UE_DRX_ENH_SW-1;
//Enabling DRX exit of UEs performing full-buffer services
MOD NRDUCELLUEPWRSAVING: NrDuCellId=0, NrDuCellDrxAlgoSwitch=FULL_BUFFER_UE_EXIT_DRX_SW-1;
//Enabling DRX status evaluation based on scheduling feedback over the PUSCH
MOD NRDUCELLUEPWRSAVING: NrDuCellId=0,
NrDuCellDrxAlgoSwitch=UL_BASED_DRX_STATUS_DETECT_SW-1;
//Configuring the DRX sleep time detection threshold
MOD NRDUCELLUEPWRSAVING: NrDuCellId=0, DrxSleepStateDetectThld=2;
//Enabling PDCCH beam robustness
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoSwitch=PDCCH_BEAM_ROBUST_SW-1;
//Enabling DRX configuration procedure optimization
MOD NRDUCELLUEPWRSAVING: NrDuCellId=0, NrDuCellDrxAlgoSwitch=DRX_CONFIG_OPT_SW-1;
//Enabling optimized collaboration between DRX and BWP2 in low-frequency scenarios
MOD NRDUCELLUEPWRSAVING: NrDuCellId=0, BwpPwrSavingSw=DRX_BWP2_OPT_SW-1;
//Enabling non-overlapping between CSI and SR reporting slots in low-frequency scenarios
MOD NRDUCELLUEPWRSAVING: NrDuCellId=0,
NrDuCellDrxAlgoSwitch=DRX_CSI_SLOT_SHIFT_FOR_SR_SW-1;
//Enabling downlink enhancement for power saving for low-frequency TDD cells served by the MetaAAU
MOD NRDUCELLPDSCH: NrDuCellId=0, DLLinkAdaptEnhancementSw=PWR_SAVING_DOWNLINK_ENH_SW-1;
//Enabling coexistence of DRX and gap-assisted measurement in low-frequency scenarios
MOD NRDUCELLUEPWRSAVING: NrDuCellId=0,
NrDuCellDrxAlgoSwitch=DRX_AND_GAP_MEAS_COEXIST_SW-1;
//Enabling the function of exit of weak-coverage UEs from DRX in low-frequency scenarios
MOD NRDUCELLUEPWRSAVING: NrDuCellId=0, NrDuCellDrxAlgoSwitch=WEAK_COV_DRX_EXIT_SW-1;
//Configuring the initial DMRS RSRP weak coverage threshold and initial DMRS SINR weak coverage
threshold
MOD NRDUCELLDMRS: NrDuCellId=0, InitDmrsRsrpWeakCovThld=-130, InitDmrsSinrWeakCovThld=0;
//Configuring the weak coverage threshold offset, weak coverage CQI threshold, and weak coverage uplink
SINR threshold
MOD NRDUCELLUEPWRSAVING: NrDuCellId=0, WeakCoverageThldOffset=5, WeakCoverageCqiThld=5,
WeakCoverageUlsinrThld=5;
//Enabling the inactivity timer length adaptation function in inter-gNodeB CA scenarios
MOD NRDUCELLUEPWRSAVING: NrDuCellId=0, IAAadaptiveAlgoSwitch=ON_IPRAN_CA;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling the DRX function for a specific QCI
MOD NRCELLQCIBEARER: NrCellId=0, Qci=100, DrxParamGroupId=255;
MOD NRDUCELLQCIBEARER: NrDuCellId=0, Qci=100, DuDrxParamGroupId=255;
//Disabling the DRX function for a cell
MOD NRDUCELLUEPWRSAVING: NrDuCellId=0, NrDuCellDrxAlgoSwitch=BASIC_DRX_SW-0;
```


4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.2 Activation Verification

Prerequisites: The UE supports DRX.

The activation verification procedure is as follows:

1. Run the **MOD NRDUCELLUEPWRSAVING** command with the **BASIC_DRX_SW** option of the **NRDUCellUePwrSaving.NrDuCellDrxAlgoSwitch** parameter selected to activate DRX.
2. Enable a UE to access the network. Check the RRCConnectionReconfiguration message (in NSA networking) or the RRCReconfiguration message (in SA networking) over the Uu interface. If the message contains DRX parameters, DRX has taken effect.

If it is observed that the DRX Inactivity Timer length is not consistent with the **gNBDUDrxParamGroup.DrxInactivityTimer** parameter value, it indicates that the inactivity timer length adaptation function has taken effect in inter-gNodeB CA scenarios.

NOTE

In low-frequency scenarios, to ensure that a UE has the opportunity to send periodic SRS and periodic CSI feedback within the length of On Duration Timer, the gNodeB may extend the length. As a result, the On Duration Timer length in effect will be inconsistent with the value of the **gNBDUDrxParamGroup.OnDurationTimer** parameter configured for the gNodeB. If the extended On Duration Timer length exceeds the long DRX cycle, the gNodeB does not deliver the DRX configuration to the UE. That is, the *DRX-Config* field is not filled in the RRCConnectionReconfiguration message (in NSA networking) or the RRCReconfiguration message (in SA networking) delivered by the gNodeB to the UE.

4.4.3 Network Monitoring

After DRX is enabled, monitor the following counters to evaluate the impact of DRX on delay and downlink throughput. For details about the impact, see [4.2.2 Impacts](#).

- Compare the results of **(N.QoS.DL.PktDelaygNBDU.Time/N.QoS.DL.PktDelaygNBDU.Num) + (N.QoS.DL.PktDelayAirInterface.Time/N.QoS.DL.PktDelayAirInterface.Num)** before and after DRX activation and evaluate the impact of DRX on service delay.
- Check the result of **N.ThpVol.DL.Cell/N.ThpTime.DL.Cell** to evaluate the impact of DRX on throughput.
- Check the result of **N.Cdrx.Sleep.Dur.Total** (total duration of UEs in the sleep time in a cell) and **N.Cdrx.Active.Dur.Total** (total duration of UEs in the active time in a cell) to evaluate the power saving effect of UEs.

Table 4-11 describes the mapping between the IDs and names of the preceding counters.

Table 4-11 Mapping between counter IDs and counter names

Counter ID	Counter Name
1911816924	N.QoS.DL.PktDelaygNBDU.Time
1911816925	N.QoS.DL.PktDelayAirInterface.Num
1911816926	N.QoS.DL.PktDelayAirInterface.Time
1911816927	N.QoS.DL.PktDelaygNBDU.Num
1911816694	N.ThpVol.DL.Cell
1911816695	N.ThpTime.DL.Cell
1911820769	N.Cdrx.Sleep.Dur.Total
1911820770	N.Cdrx.Active.Dur.Total

5 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

6 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

- Step 1** Open the EXCEL file of performance counter reference.
- Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All counters related to the feature are displayed.

----End

7 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

8 Reference Documents

- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- 3GPP TS 38.321: "NR; Medium Access Control (MAC) protocol specification"
- 3GPP TS 38.331: "NR; Radio Resource Control (RRC) protocol specification"
- *ANR*
- *Channel Management*
- *Uplink Scheduling*
- *QoS Management*
- *Flexible User Steering*
- *EPC-based NSA Performance Enhancement*
- *UE Power Saving*
- *VoNR*
- *High Speed Mobility*
- *CoMP*
- *Carrier Aggregation*
- *URLLC*
- *Co-MM (Low-Frequency TDD)*
- *Positioning*
- *Remote Interference Management (Low-Frequency TDD)*
- *Coordinated Interference Management (Low-Frequency TDD)*
- *mmWave Beam Management (High-Frequency TDD)*
- *5G Networking and Signaling*
- *Network Slicing*
- *CA Experience Improvement*
- *FWA*
- *Service-differentiated Experience Guarantee*
- *Massive MIMO iBeam (Low-Frequency TDD)*
- *Beam Management*
- *Technical Specifications in 3900 & 5900 Series Base Station Product Documentation*